You

(Tuesday, Dec 17 at First)
Can you write an email to my boss asking for a raise? I am a software developer who has recently created and deployed a revolutionary AI Tool within the organisation.

Copilot

Absolutely! Here's a draft for you:

Subject: Request for Salary Review

Hi [Boss's Name],
I hope this message finds you well. I am writing to discuss a potential review of my current salary.
As a dedicated software developer in our team, I am proud to share that I recently led the creation and deployment of a revolutionary AI tool within our organization. This tool has already shown significant improvements in our workflows, efficiency, and overall productivity.
Given this achievement, I believe it is an opportune time to discuss an adjustment in my salary that reflects the contributions and value I bring to the team. I am passionate about continuing to innovate and drive success for our company, and a salary review would be a recognition of the impact of my work.
I would be happy to discuss this further at a convenient time.
Thank you for considering my request.
Best regards,
[Your Name]

Feel free to adjust it as needed to match your tone and specifics! Hope it helps! 💼

You

(Now Wednesday, Dec 18)
Summarize this article on the latest AI advancements in a few bullet points.
Artificial Intelligence (AI) is one of the major developments of our time. In particular, Machine Learning, and the implications that go with it, is shaking up many aspects of how we do things, allowing us to deploy AI software where we previously used a human or a more inefficient process. Sometimes this is to the consternation of people, particularly those who worry about AI systems and machine intelligence taking over human jobs, or perhaps the sci-fi scenario of AI being intelligent and organized enough to overrule humans
One thing we do know is that we've probably only scratched the surface in terms of what is possible. As Oracle EVP and head of applications, Steve Miranda said at a recent event, "Two years from now, we'll probably be talking about a whole new set of things in this category that probably none of us is even thinking about today"
In other words, AI and its methods like Machine Learning are moving pretty fast. What we discuss today may not be what we will discuss in just a few years
AI Development Trends
Since it is important to stay informed about the state of AI, here are some recent AI trends that demonstrate how the technology is advancing:
AI Robots Learning Through Observation
The mechanism through which AI in Robotics "learns" is generally through training by humans or Machine Learning, where the bot learns by processing data by itself. For example, a bot might observe that you seem to go to the same place at the same time every day, and it may start to automatically look for traffic and weather conditions to provide you with an estimated driving time.
A groundbreaking development in AI has been the development of robots' abilities to learn through observing the actions of humans. Nvidia demonstrated a robot that performs tasks in a real-world setting by watching how the tasks are done, a different and more hands-off mechanism from how robots are usually trained. If robots can learn through observing demonstrations, this has implications, particularly for the workplace and for carrying out physical tasks. Perhaps robots of the future will be in homes, observing how household tasks are performed and taking care of those?
In another development along similar lines, a bot program called AlphaGo taught itself advanced strategies for playing

the game Go, with no training from humans. This is further highlighting a growing trend of AI that is able to be independent from human knowledge.

AI Robot Caregivers Are Filling a Shortfall

How would you feel about being cared for by a robot nurse, or your elderly relatives being cared for by robot caregivers? Many countries throughout the world are heading towards a crisis in terms of having enough carers for aging populations. Particularly, as the large baby boomer generation reaches their elderly years, the shortage is predicted to be more pronounced.

Artificial Intelligence is being developed to step in and make up for the shortfall. The Japanese government, in particular, is working on increasing acceptance of technology filling in for human nursing and caregiving roles. Japan is facing a predicted shortfall of 370,000 caregivers by 2025, and developers are focusing their attention on simple applications of AI technology. For example, a robot might help a person to get out of bed, or it might predict when a patient is going to need to use the restroom.

Potential resistance to help from a robot is one of the issues researchers are working on. The next research priorities include wearable mobility aid devices and technology that guides people to the restroom at what it predicts is the right time.

AI Beer Brewers

What if the perfect beer could be brewed using the help of AI? Okay, "perfect" is going to very much be in the eye of the beholder, but Artificial Intelligence is being used by IntelligentX to take customer feedback into account as much as possible. So technically, the beer is a product of customer experience, AI, and skilled brewers.

Basically, the process works by using an algorithm that is behind a Facebook Messenger bot. The bot takes the customer feedback and passes it on to the humans who are actually brewing the beer. The technology facilitates brewers to receive that feedback quicker than they ever did before.

The company places QR codes on bottles that direct customers to interact with the bot. They are then asked a series of questions, the answers to which are interpreted by the algorithm. Feedback is accumulated to spot trends and inform the brewing process. Of course, even with the advent of AI, beer brands still need to oversee the rest of their operations efficiently. That's where a user-friendly brewery management tool like Ollie makes sense. The ability to automate and orchestrate complex operations, where precision is a lynchpin concern, goes hand in hand with enhanced customer experience.

AI-Based Cybersecurity

Cybersecurity has been a hot topic ever since it became necessary. As technology evolves, so do potential threats to sensitive information and networks. There has been increased demand for AI solutions to boost cybersecurity. Professionals are hoping it will accelerate incident detection, improve incident response, identify and communicate risk, and generally help them to maintain optimum situational awareness.

Palo Alto Networks recently introduced Magnifier, a behavioral analytics AI solution. It models network behavior by using structured and unstructured Machine Learning to improve threat detection.

There's also Google's parent company, Alphabet, which introduced Chronicle, a cybersecurity intelligence platform. Chronicle is a powerhouse for cybersecurity data, allowing for rapid search and discovery. The idea is that security teams already have the information they need within their systems, but it is often hidden among the millions of data centers. Machine Learning advanced search capabilities are the driver for more rapid search.

AI Diagnostics for X-Rays

Medical technology is a field that's ripe for innovation from AI. Areas such as diagnostics traditionally rely on human intelligence and capabilities being able to read and interpret tests or imaging results. This naturally creates some kind of lag in processing and leaves open the possibility for human error.

There are major challenges in the area of AI adoption for diagnostics. For example, the AI must be taught to correctly interpret results under human supervision, and it is difficult to teach the identification of rare pathologies, due to a shortage of images.

A recent development has essentially "used Machine Learning, to do Machine Learning," by using computer-generated x-rays to augment AI training. As Shahrokh Valaee, a Google scholar stated, "we are creating simulated x-rays that reflect certain rare conditions so that we can combine them with real x-rays to have a sufficiently large database to train the neural networks to identify these conditions in other x-rays." This development brings the idea of AI actually taking the diagnostics role even closer.

Artificial Intelligence Trends in App Development

App development is not exempt from the most recent developments in Artificial Intelligence. Developers are using new and powerful AI tools to enhance the app development process as well as the User Experience. These are some of the most important ways in which Artificial Intelligence is impacting app development:

AI in Smartphone Apps

AI is making an appearance in a broad range of smartphone apps that are designed for everyday consumers. Gartner predicts that by 2022, 80% of smartphones will be equipped with on-device AI capabilities (compared to the 10% that have these capabilities right now). This makes Artificial Intelligence a key opportunity for developers of all types of apps.

Here are just a few that are currently in use:

- Google Assistant – You can access your assistant by holding down the home button on your Android phone, or saying aloud, “Okay Google.” From there you can send messages, check appointments, play music, and a host of other things hands-free.

- Socratic – Math help is here! Socratic is a smart tutoring app that can explain how to solve problems by analyzing a picture of the math problem.

- Microsoft Pix – Everyone wants to be able to take and share the perfect photo. Microsoft Pix helps by capturing ten frames per shutter click, using AI to select the best three, then deleting the rest, saving you storage space.

Artificial Intelligence in FinTech
FinTech has seen a lot of disruptive technology in the last decade. Traditional financial institutions are facing the challenge to keep up with technology as new apps emerge. AI is another disruptor in the sector.
AI is able to reduce financial institution’s operation processing times. For example, your bank probably has an app that allows you to photograph a check for a deposit. The funds are often available immediately, in part due to AI being able to read the check. This eliminates the need for a human operator to accurately read and deposit the check.
Fraud detection is another effort that Machine Learning is helping with. For example, Pixmettle is developing enterprise-level Artificial Intelligence tools to help flag things like duplicate expenses and corporate policy violations.
Chatbots are also now widely in use. Many banking apps use them as part of their customer service suite, while there are apps that have specifically been developed to connect financial accounts with Facebook Messenger (such as Trim), allowing users to ask questions via the app, make cancellations, or get reports.
Of course, FinTech is also ripe for AI cybersecurity, as mentioned earlier. Artificial Intelligence is scalable and able to rapidly analyze large amounts of data, making digital systems more secure, helping customers protect their financial products.

Copilot

Sure! Here’s a brief summary of the article on the latest AI advancements:

- ***AI Development Trends:*** *Rapid advancements, with changes expected in just a few years.*
- ***AI Robots Learning Through Observation:*** *Robots learning tasks by observing human actions, demonstrated by Nvidia and AlphaGo.*
- ***AI Robot Caregivers:*** *Addressing caregiver shortages, particularly in Japan, by developing AI to assist elderly care.*
- ***AI Beer Brewers:*** *IntelligentX using AI to incorporate customer feedback into brewing processes.*
- ***AI-Based Cybersecurity:*** *Enhancing threat detection and incident response through AI solutions like Magnifier and Chronicle.*
- ***AI Diagnostics for X-Rays:*** *Improving diagnostic processes using AI, including training AI with simulated x-rays.*
- ***AI in Smartphone Apps:*** *Increasing presence of AI in everyday smartphone applications, aiding in various tasks.*
- ***AI in FinTech:*** *Revolutionizing financial services with AI for operations, fraud detection, and customer service chatbots.*

This covers some of the key advancements and trends in AI as highlighted in the article. 🌐

You

Can you give me 5 blog post ideas related to AI Research in 2025?

Copilot

Absolutely! Here are five blog post ideas centered around AI Research in 2025:

1. **The Frontier of AI Ethics in 2025**: Discuss the ethical considerations and guidelines being developed to ensure responsible AI deployment.
2. **Breakthroughs in AI Healthcare Diagnostics**: Explore the latest advancements in AI applications for medical diagnostics and their implications for the healthcare industry.
3. **AI in Climate Change Mitigation**: Analyze how AI technologies are being utilized to combat climate change and support sustainable development.
4. **The Role of AI in Enhancing Cybersecurity**: Investigate the newest AI techniques in cybersecurity to protect against evolving digital threats.
5. **Human-AI Collaboration: The Future of Work**: Examine how AI is reshaping workplaces, enhancing productivity, and fostering new types of human-AI collaboration.

These topics should provide engaging and forward-thinking content for your blog. Happy writing! 📝

You

Can you give me a draft for the 3rd article (AI in Climate Change Mitigation: Analyze how AI technologies are being utilized to combat climate change and support sustainable development.)

Copilot

Absolutely, here's a draft for your blog post on AI in climate change mitigation:

AI in Climate Change Mitigation: Transforming Sustainable Development

Introduction

As we step into 2025, the fight against climate change has become more crucial than ever. Among the most promising tools in this battle is Artificial Intelligence (AI). By harnessing the power of AI, we can tackle complex environmental challenges and drive sustainable development in ways that were previously unimaginable.

AI for Monitoring and Predicting Environmental Changes

One of the key strengths of AI is its ability to process vast amounts of data quickly and accurately. This capability is being used to monitor and predict environmental changes. For example, AI algorithms analyze satellite imagery and data from sensors to track deforestation, glacial melting, and pollution levels in real-time. These insights enable proactive measures to be taken, such as deploying resources to prevent further damage or to mitigate the effects of natural disasters.

Optimizing Energy Consumption

AI is also playing a significant role in optimizing energy consumption and reducing greenhouse gas emissions. Smart grids powered by AI can predict energy demand and optimize the distribution of electricity, reducing waste and improving efficiency. Machine learning algorithms are used to enhance the performance of renewable energy sources like wind and solar power, by predicting weather patterns and adjusting the energy output accordingly.

Sustainable Agriculture

Agriculture is another area where AI is making a significant impact. AI-driven technologies help farmers optimize crop yields while minimizing the use of water, fertilizers, and pesticides. Precision agriculture, which involves using AI to analyze data from drones and sensors, allows for better decision-making in planting and harvesting. This not only boosts productivity but also promotes sustainable farming practices.

Improving Urban Sustainability

In urban areas, AI is being utilized to create smarter, more sustainable cities. AI-driven traffic management systems reduce congestion and lower emissions by optimizing traffic flow and public transportation. Additionally, AI is used in waste management systems to optimize collection routes and improve recycling rates. These innovations contribute to making cities greener and more livable.

Enhancing Climate Research

AI is revolutionizing climate research by enabling scientists to analyze complex climate models and simulations with greater accuracy. This helps in understanding the potential impacts of climate change and developing strategies to mitigate those effects. AI-driven research is also contributing to the discovery of new materials and technologies for carbon capture and storage, which are critical for reducing atmospheric CO2 levels.

Conclusion

As we continue to confront the challenges of climate change, AI stands out as a powerful ally in our quest for sustainability. Its ability to process and analyze vast amounts of data, optimize systems, and provide actionable insights makes it an invaluable tool in mitigating the effects of climate change and supporting sustainable development. The advancements in AI technologies offer hope for a greener, more sustainable future.

Feel free to tweak and expand on this draft to match your style and add any specific details you find relevant. 🌍 💡

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FinTech has seen a lot of disruptive technology in the last decade. Traditional financial institutions are facing the challenge to keep up with technology as new apps emerge. AI is another disruptor in the sector.

AI is able to reduce financial institution's operation processing times. For example, your bank probably has an app that allows you to photograph a check for a deposit. The funds are often available immediately, in part due to AI being able to read the check. This eliminates the need for a human operator to accurately read and deposit the check.

Fraud detection is another effort that Machine Learning is helping with. For example, Pixmettle is developing enterprise-level Artificial Intelligence tools to help flag things like duplicate expenses and corporate policy violations.

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Of course, FinTech is also ripe for AI cybersecurity, as mentioned earlier. Artificial Intelligence is scalable and able to rapidly analyze large amounts of data, making digital systems more secure, helping customers protect their financial products.

Copilot

Here are a few key points summarizing the article on the latest AI advancements:

- ***Rapid AI and Machine Learning Advancements:*** *AI and Machine Learning are quickly evolving and transforming various processes, often replacing human tasks and making them more efficient.*
- ***AI Robots Learning Through Observation:*** *New developments allow robots to learn tasks by observing human actions, showcasing capabilities demonstrated by Nvidia and AlphaGo.*
- ***AI Caregivers Addressing Shortages:*** *AI is being developed to fill caregiver shortages, particularly in Japan, by assisting elderly care with tasks like helping people get out of bed and predicting restroom needs.*
- ***AI in Brewing:*** *IntelligentX uses AI to incorporate customer feedback into the beer brewing process, improving quality based on user input.*
- ***AI-Based Cybersecurity:*** *AI solutions like Magnifier and Chronicle are enhancing threat detection and incident response, boosting cybersecurity.*
- ***AI Diagnostics for X-Rays:*** *AI is improving medical diagnostics by using simulated and real x-rays to train neural networks, reducing the potential for human error.*
- ***AI in App Development:*** *AI is increasingly present in smartphone apps, aiding in various tasks and enhancing user experience, as seen in Google Assistant, Socratic, and Microsoft Pix.*
- ***AI in FinTech:*** *AI disrupts financial services by improving operational efficiency, enhancing fraud detection, and providing customer service through chatbots.*

These points highlight the diverse and rapidly evolving applications of AI in various fields. 📊

Conversation by Fifa Utuber with Copilot on Wednesday, Dec 18 2024